TRAFFIC LIGHT CONTROLLER

DESIGN MODULE IMPLEMENTATION (VERILOG HDL)

```
module traffic_light (
    input clk,
    input rst,
    output reg AG, AY, AR,
    output reg BG, BY, BR
);

parameter S0 = 2'b00;
parameter S1 = 2'b01;
parameter S2 = 2'b10;
parameter S3 = 2'b11;
parameter DELAY = 10;

reg [1:0] state;
reg [3:0] count;

always @(posedge clk or posedge rst)
begin
    if (rst) begin
        state <= S0;
        count <= 0;
    end else begin
        if (count == DELAY - 1) begin
```

```
always @(*) begin
    AG = 0;
    AY = 0;
    AR = 0;
    BG = 0;
    BY = 0;
    BR = 0;

    case (state)
        S0: begin
            AG = 1;
            BR = 1;
        end
        S1: begin
            AY = 1;
            BR = 1;
        end

        S2: begin
            AR = 1;
            BG = 1;
        end
        S3: begin
```

```
            count <= 0;
            case (state)
                S0: state <= S1;
                S1: state <= S2;
                S2: state <= S3;
                S3: state <= S0;
                default: state <= S0;
            endcase
        end else begin
            count <= count + 1;
        end
    end
end
```

```
                AR = 1;
                BY = 1;
            end
            default: begin
                AG = 1;
                BR = 1;
            end
        endcase
    end

endmodule
```

TESTBENCH:

```
module tb_traffic_light;
reg clk;
reg rst;
wire AG, AY, AR;
wire BG, BY, BR;

traffic_light dut (
    .clk(clk),
    .rst(rst),
    .AG(AG),
    .AY(AY),
```

```
// Clock generation: 10 ns period
initial begin
    clk = 0;
    forever #5 clk = ~clk;
end

// Reset and simulation
initial begin
    $dumpfile("traffic_light.vcd");
    $dumpvars(0, tb_traffic_light);
    $monitor(
```

```
    .AR(AR),
    .BG(BG),
    .BY(BY),
    .BR(BR)
);
```

```
        "Time=%0t rst=%b | A: G=%b Y=%b R=%b | B: G=%b Y=%b R=%b",
        $time, rst, AG, AY, AR, BG, BY, BR
    );
    rst = 1;
    #12;
    rst = 0;
    #450;
    $finish;
end
endmodule
```

